

**JDSE 2026 • ORAL
PRESENTATION**

Direct Multimodal Clinical Information Extraction

from digitized patient records, with Gold Standard evaluation in sepsis

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Clinical motivation



49 million

sepsis cases worldwide in 2017



11 million deaths

Nearly one in five deaths globally was associated
with sepsis

The information required for early detection is distributed across multiple document types

- Clinical observations and ICU reports
- Laboratory and microbiology results
- Medical imaging, prescriptions, and therapeutic information
- Scanned records with variable visual quality and complex layouts

Clinical value depends on converting this fragmented evidence into structured knowledge.

Research gap and question



OCR errors can propagate to every downstream step

Research question

Can multimodal language models extract clinical entities and relations directly from record images, without an intermediate OCR transcription step?

This study explores direct extraction. An OCR+LLM comparison remains future work.

[1] KG-EDS pipeline

Study contributions

Direct multimodal extraction

A complete image-to-JSON pipeline for clinical entities and relations.

Human-annotated Gold Standard

A Label Studio reference corpus guided by the TRACE-Sepsis annotation protocol.

Controlled model comparison

Qwen2.5-VL-7B-Instruct and Mistral Small 4 evaluated under equivalent conditions.

Current extension

The corpus expanded from 211 records at submission to 246 records after submission.

Corpus and evaluation scope



Corpus at
submission
211

Current corpus:
264



Gold Standard records

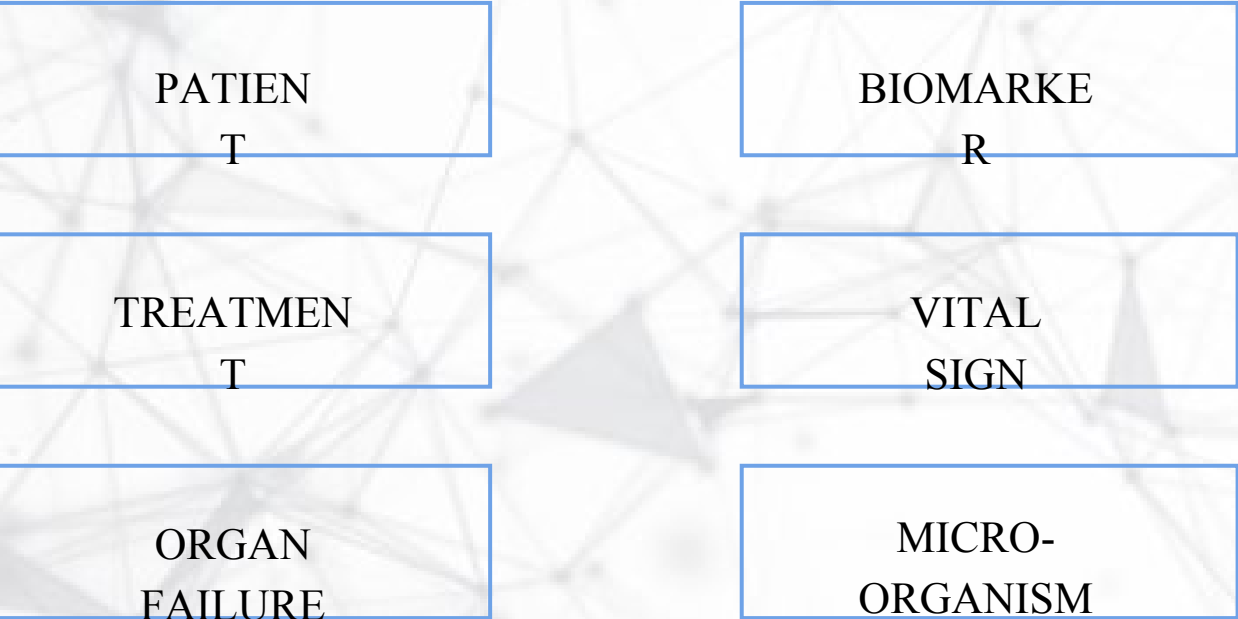
4 $\approx$ **20**
3 %

Document content

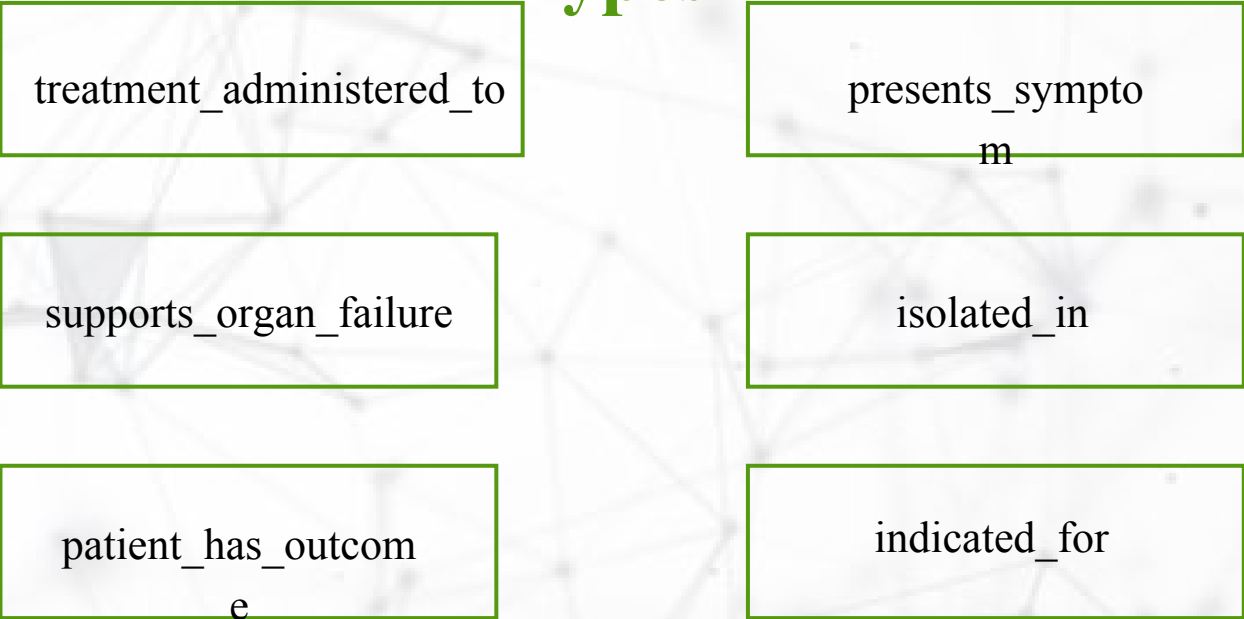
- Hospitalization and intensive care reports
- Clinical observations
- Biological and microbiological results
- Prescriptions and therapeutic information

TRACE-Sepsis Annotation Schema and Guidelines

21 entity types



32 relation types



The same terminology constrains model output and guides reference annotation.

TRACE-Sepsis Annotation Schema and Guidelines

A standardized framework for consistent clinical information extraction

Clinical ambiguity

“Fever at 39°C”

Should ‘fever’ alone be annotated?

Should the temperature be included?

Which clinical category applies?

Without explicit rules, annotators may represent the same evidence differently.

TRACE-Sepsis specifies

1

Category definition

2

Inclusion and exclusion criteria

3

Annotation span and granularity

4

Management of ambiguous cases

5

Authorized entity relations

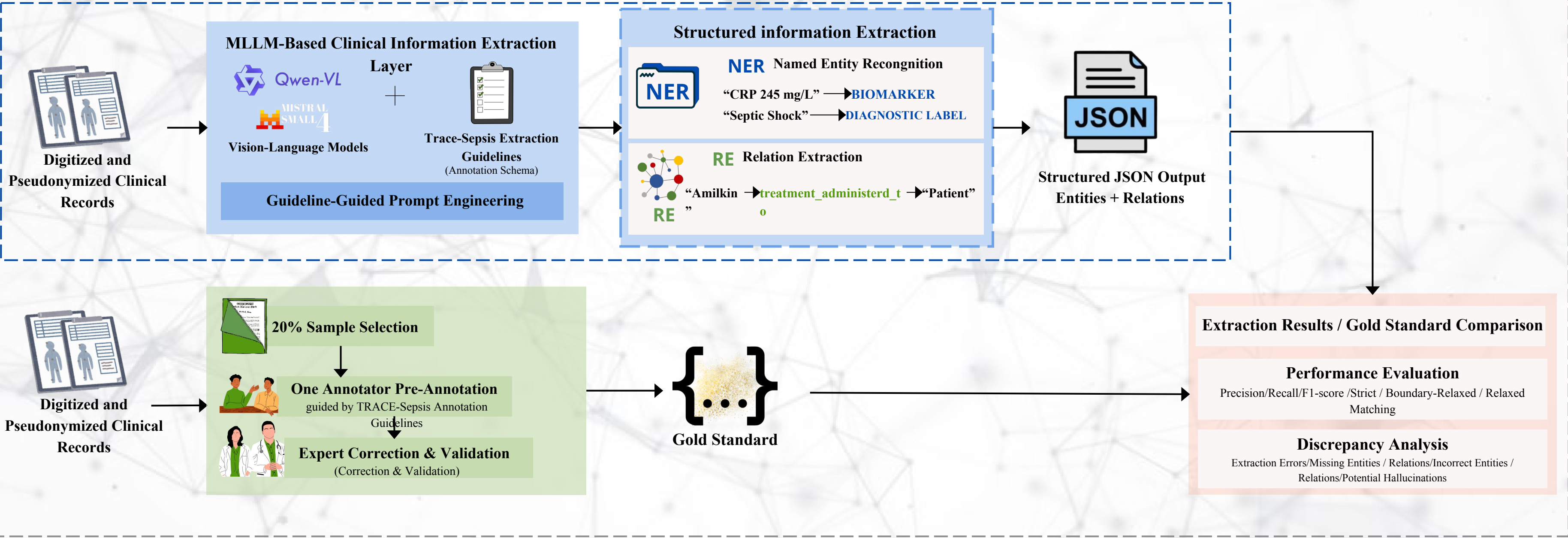
Standardized decision

Clinical evidence	Assigned category
Type 2 diabetes	COMORBIDITY_HISTORY
Septic shock	DIAGNOSTIC_LABEL
Pneumonia	DIAGNOSTIC_LABEL

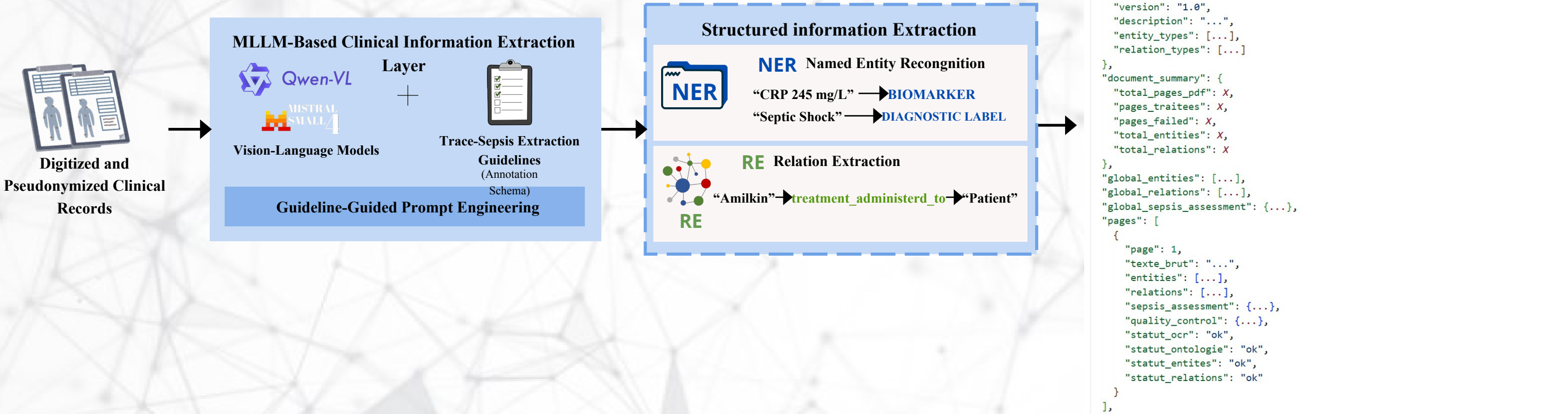
The same rules guide human annotation and model extraction.

Direct multimodal extraction pipeline

Work Presented In The Study



Direct multimodal extraction pipeline



Structured JSON Output: Entities and Relations

```

{
  "source_file": "nom.pdf",
  "type_extraction": "lecture_visuelle_directe_plus_schema_trace_sepsis_detailed",
  "schema": {
    "name": "schema_trace_sepsis_fr",
    "version": "1.0",
    "description": "...",
    "entity_types": [...],
    "relation_types": [...]
  },
  "document_summary": {
    "total_pages_pdf": X,
    "pages_traitees": X,
    "pages_failed": X,
    "total_entities": X,
    "total_relations": X
  },
  "global_entities": [...],
  "global_relations": [...],
  "global_sepsis_assessment": {...},
  "pages": [
    {
      "page": 1,
      "texte_brut": "...",
      "entities": [...],
      "relations": [...],
      "sepsis_assessment": {...},
      "quality_control": {...},
      "statut_ocr": "ok",
      "statut_ontologie": "ok",
      "statut_entities": "ok",
      "statut_relations": "ok"
    }
  ],

```

```

{
  "page": 3,
  "entites": [
    {
      "identifiant_entite": "E001",
      "categorie": "BIOMARQUEUR",
      "parametre": "proteine_C_reactive",
      "valeur": 97.2,
      "unite": "mg/L",
      "valeur_reference": null,
      "direction_cinetique": "inconnue",
      "horodatage": "inconnu",
      "preuve": "CRP 97.2 mg/L",
      "nie": false,
      "confiance": "elevee",
      "type_inference": "extraction_directe"
    }
  ]
}

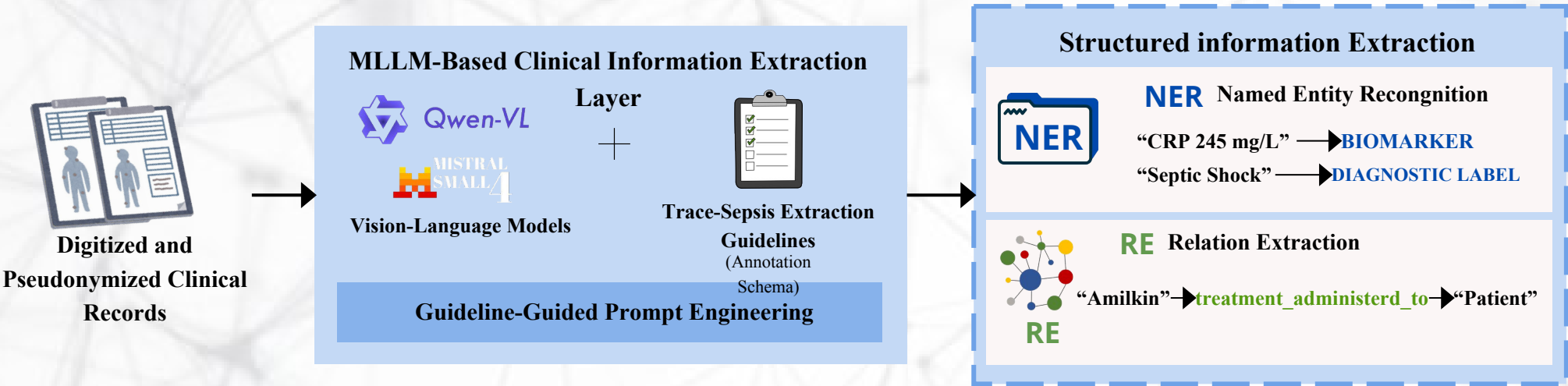
```

```

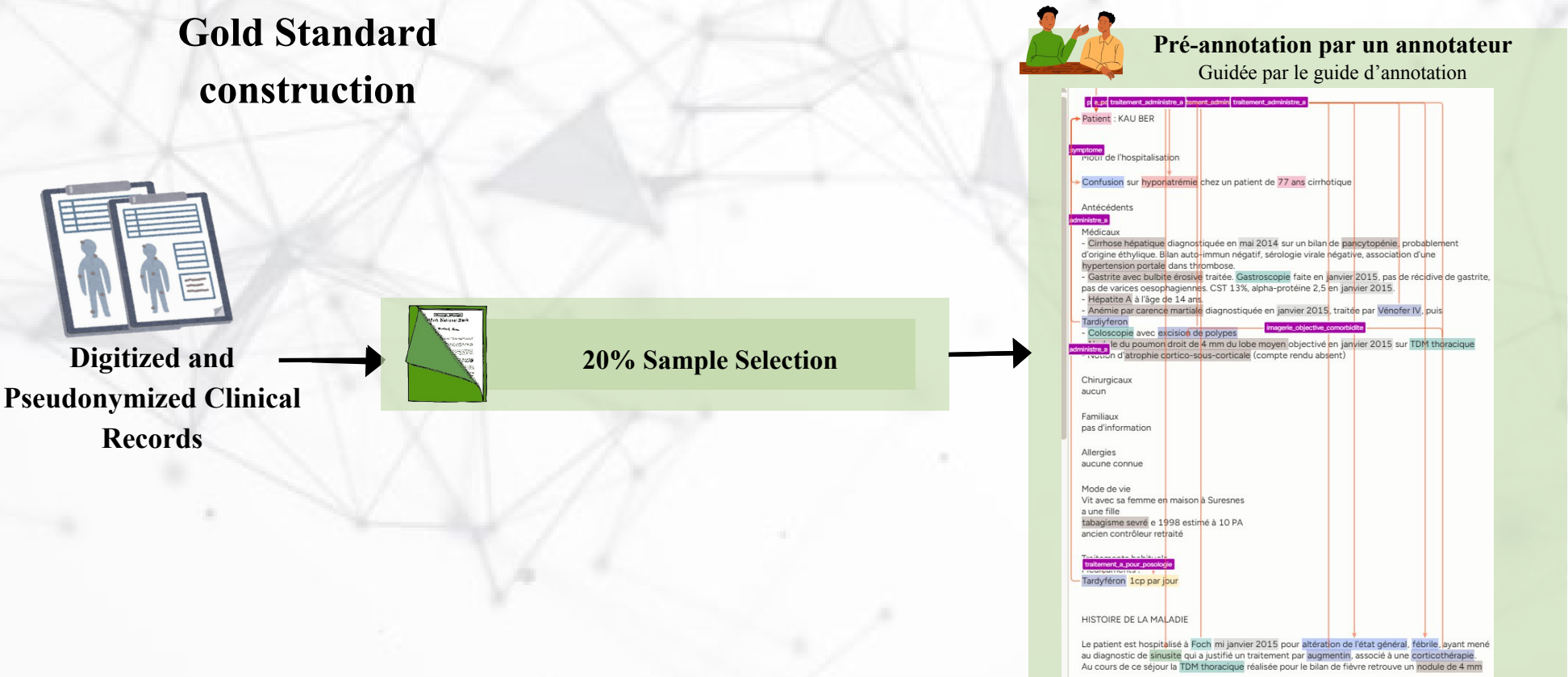
{
  "page": 3,
  "relations": [
    {
      "identifiant_relation": "R001",
      "name_entite_sujet": "Amiklin",
      "type_relation": "traitement_a_pour_posologie",
      "name_entite_objet": "5 jours",
      "dela_minutes": null,
      "preuve": "Amiklin pendant 5 jours",
      "note_clinique": "",
      "type_inference": "extraction_directe",
      "confiance": "elevee"
    }
  ]
}

```


Direct multimodal extraction pipeline



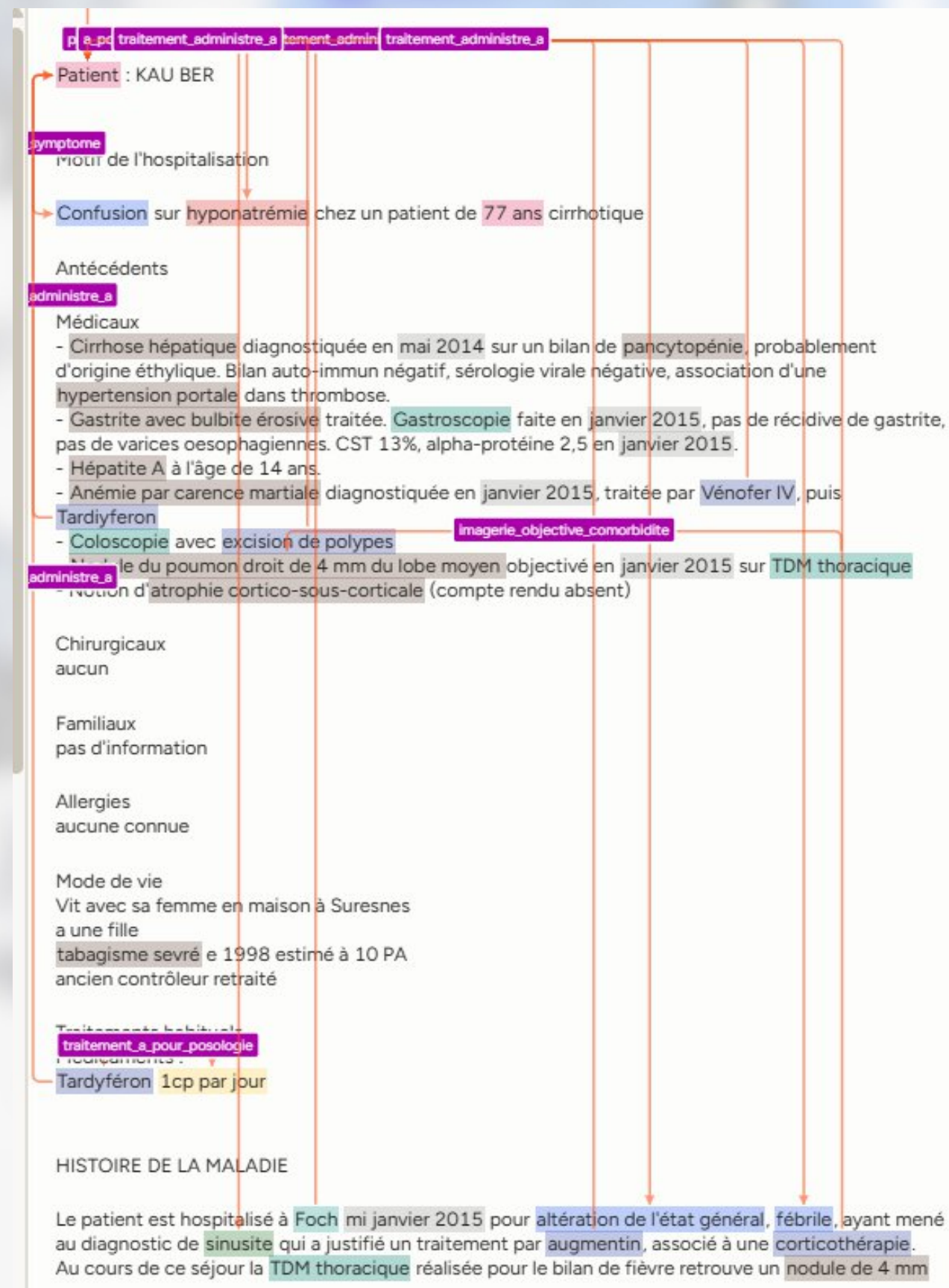
Gold Standard construction



```
{
  "source_file": "nom.pdf",
  "type_extraction": "lecture_visuelle_directe_plus_schema_trace_sepsis_detailed",
  "schema": {
    "name": "schema_trace_sepsis_fr",
    "version": "1.0",
    "description": "...",
    "entity_types": [...],
    "relation_types": [...]
  },
  "document_summary": {
    "total_pages_pdf": X,
    "pages_traitees": X,
    "pages_failed": X,
    "total_entities": X,
    "total_relations": X
  },
  "global_entities": [...],
  "global_relations": [...],
  "global_sepsis_assessment": {...},
  "pages": [
    {
      "page": 1,
      "texte_brut": "...",
      "entities": [...],
      "relations": [...],
      "sepsis_assessment": {...},
      "quality_control": {...},
      "statut_ocr": "ok",
      "statut_ontologie": "ok",
      "statut_entites": "ok",
      "statut_relations": "ok"
    }
  ]
}
```

Structured JSON Output: Entities and Relations

 One Annotator Pre-Annotation
guided by TRACE-Sepsis Annotation Guidelines



• **Entities**

DONNEE_PATIENT 1	LABEL_NOSOLOGIQUE 2	SIGNE_VITAL 3	BIOMARQUEUR 4
SCORE_SOFA 5	SCORE_qSOFA 6	SCORE_NEUROLOGIQUE 7	STADE_IRA 8
FOYER_INFECTIEUX 9	MICRO_ORGANISME 0	DEFAILLANCE_ORGANE q	
TRAITEMENT w	POSOLOGIE e	CONTEXTE_ACQUISITION t	
COMORBIDITE_ANTECEDENT a	EVENEMENT_TEMPOREL s		
EVOLUTION_PRONOSTIC d	SERVICE_MEDICAL f	IMAGERIE_PROCEDURE g	
METADONNEES_PIPELINE z	SYMPTOME x		

• **Relations**

```
<Relations>

<!-- Diagnostic -->
<Relation value="a_pour_label_nosologique"/>
<Relation value="presente_suspicion_infection"/>
<Relation value="presente_dysfonction_organe"/>

<Relation value="score_sofa_calcule_pour"/>
<Relation value="score_sofa_inclut_biomarqueur"/>
<Relation value="score_sofa_inclut_signe_vital"/>
<Relation value="score_sofa_inclut_score_neurologique"/>

<Relation value="score_qsofa_calcule_pour"/>
<Relation value="score_qsofa_inclut_score_neurologique"/>

<Relation value="stade_ira_calcule_pour"/>

<!-- Thérapeutique -->
<Relation value="traitement_administre_a"/>
<Relation value="traitement_a_pour_posologie"/>
<Relation value="antibiotique_adequat_pour"/>
<Relation value="antibiotique_inadequat_pour"/>
<Relation value="traitement_cible_defaillance"/>
<Relation value="traitement_indique_par_label_nosologique"/>

<!-- Biologie -->
<Relation value="biomarqueur_obtenu_a"/>
<Relation value="biomarqueur_supporte_defaillance_organe"/>
<Relation value="signe_vital_supporte_defaillance_organe"/>
<Relation value="biomarqueur_est_critere_de"/>
<Relation value="micro_organisme_isole_dans"/>
```

```
<!-- Terrain -->
<Relation value="comorbidite_est_facteur_risque_de"/>
<Relation value="a_pour_contexte_acquisition"/>
<Relation value="contexte_acquisition_influence_risque_resistance"/>

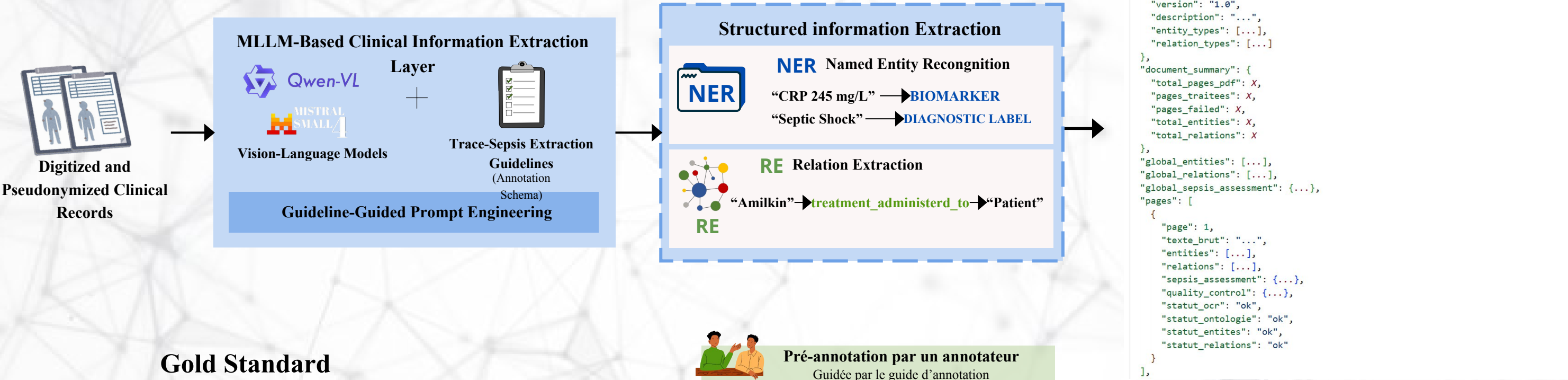
<!-- Temporel -->
<Relation value="precede"/>
<Relation value="patient_a_pour_evolution"/>
<Relation value="evenement_associe_evolution"/>
<Relation value="service_medical_accueille"/>

<!-- Imagerie -->
<Relation value="imagerie_objective_foyer"/>
<Relation value="imagerie_objective_defaillance"/>
<Relation value="imagerie_objective_comorbidite"/>

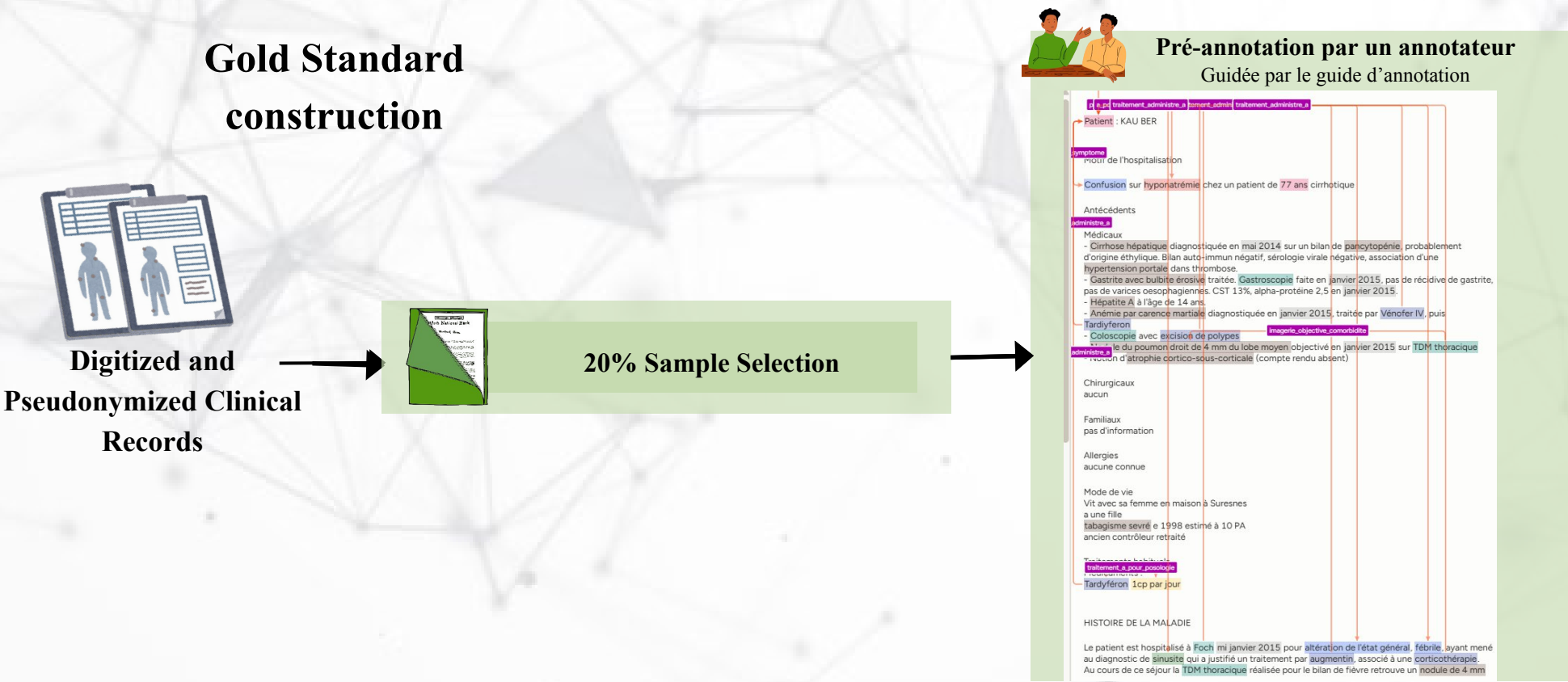
<Relation value="presente_symptome"/>
```

</Relations>

Direct multimodal extraction pipeline



Gold Standard construction



Comparaison des résultats d'extraction avec le Gold Standard

Évaluation des performances
Precision/Recall/F1-score /Strict / Boundary-Relaxed / Relaxed Matching

Analyse des écarts
Erreurs d'extraction, Entités / relations manquantes, Entités / relations incorrectes, Hallucinations potentielles

Evaluation protocol

Extraction Results / Gold Standard Comparison

Performance Evaluation

Precision/Recall/F1-score /Strict / Boundary-Relaxed / Relaxed
Matching

Discrepancy Analysis

Extraction Errors/Missing Entities / Relations/Incorrect Entities /
Relations/Potential Hallucinations

1. How do we evaluate automatic extraction ?

System Output



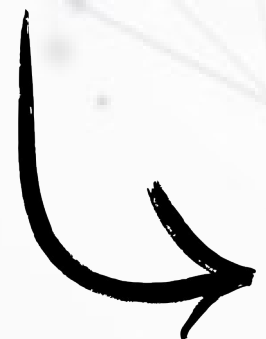
Validated Medical
Reference

2. We can identify:

- Correctly extracted information
- Missing information
- Incorrectly added information
- Incorrect category
- Incorrect relation

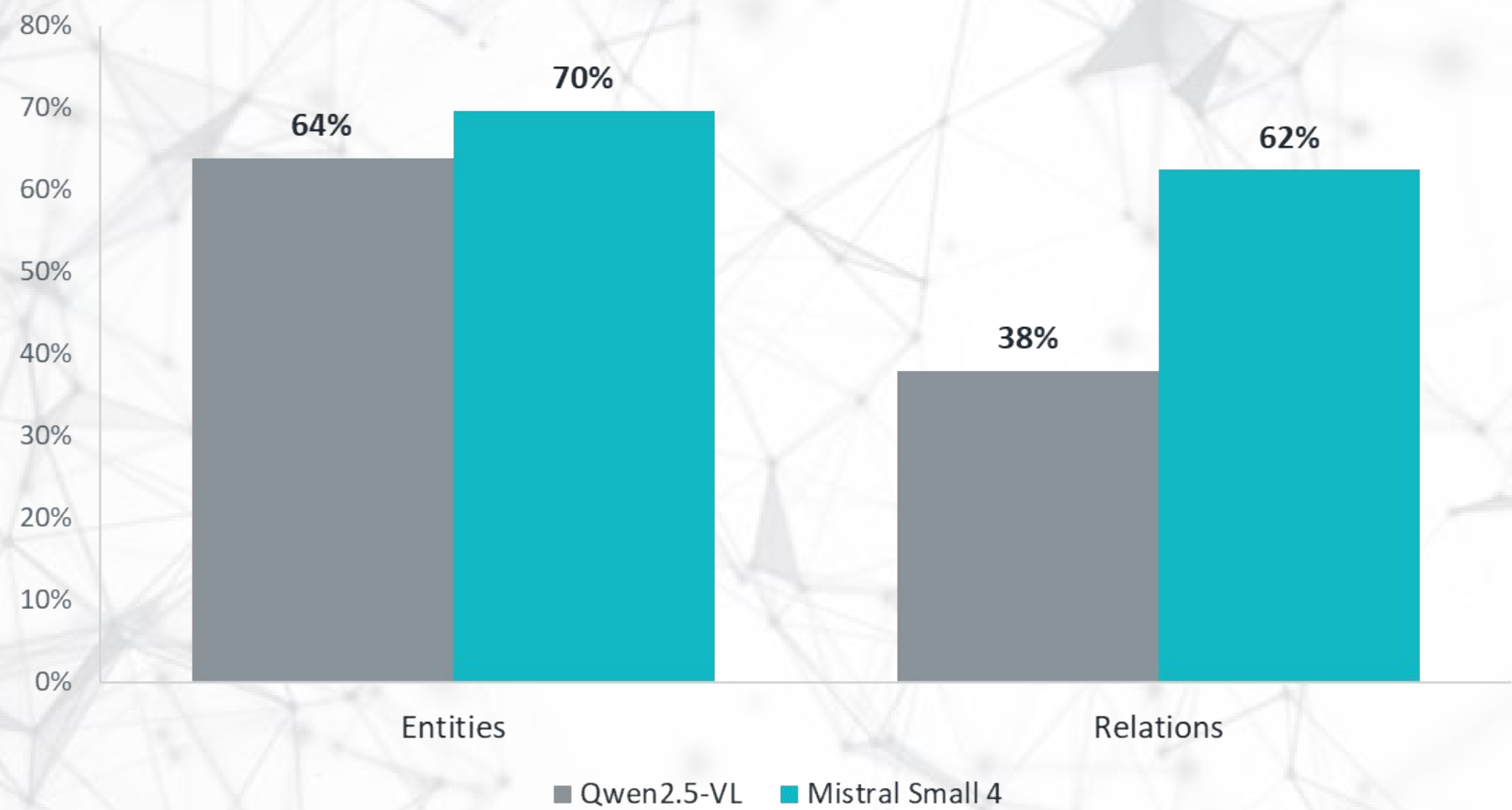
3. We then calculate:

Precision, Recall, and F1-score



The medical reference allows us to objectively assess the quality of the extracted information.

Results reported in the accepted paper



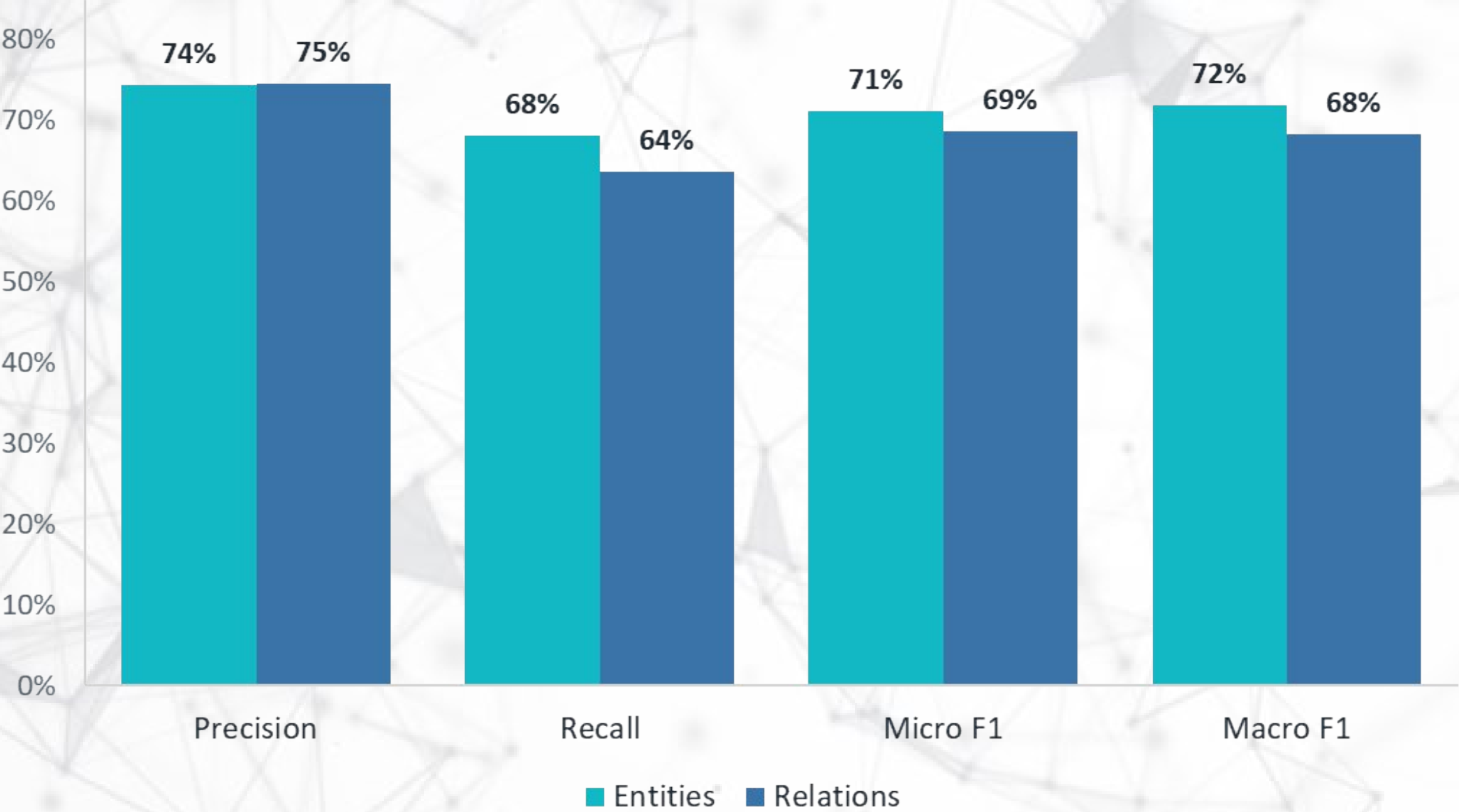
+24.45 points
**Entity and relation F1
advantage for Mistral**

Mistral Small 4

Entity F1:
69.57%
Relation F1: 62.43%

Evaluation subset: 43 annotated records

Post-submission evaluation on the same 43 records



Updated Micro F1

71.14%

Entities

68.64%

Relations

For relations

+6.21 points

Evaluation subset: 43 annotated records

Mistral Small

4

Error analysis

Omitted entities

Treatments and biomarkers may be missed when evidence is visually dense or spread across pages.

Over-extraction

Models sometimes generate plausible clinical concepts that the source document does not support.

Missing relations

Relations such as `medical_service_admits` and `presents_symptom` remain difficult to recover.

Incorrect links

The model may identify both entities correctly but connect them with the wrong relation.

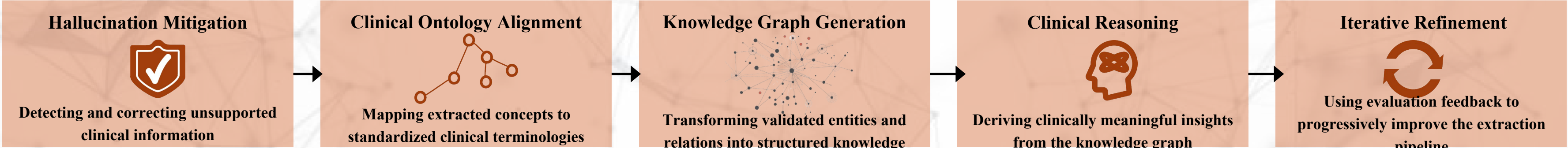
From evaluation to hallucination reduction

Hallucination taxonomy

Visual · Structural · Factual · Contextual · Temporal · Ontological · Reasoning

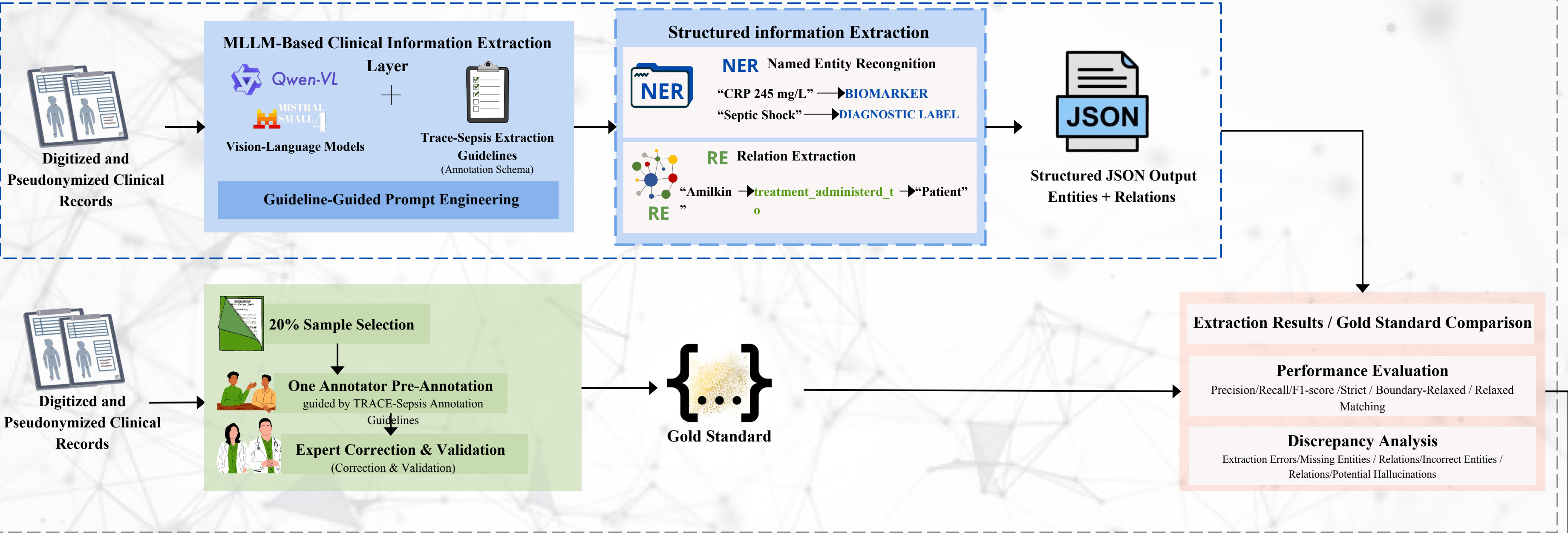
Goal: preserve useful corrections while preventing unsupported additions.

Future Work



Thank
you .
Question
s?

Work Presented In The Study



Future Work

